

COL774
Machine Learning
Aug 29, 2025

Linear Regression
Logistic Regression

Last Pass-

Generalized Linear Models

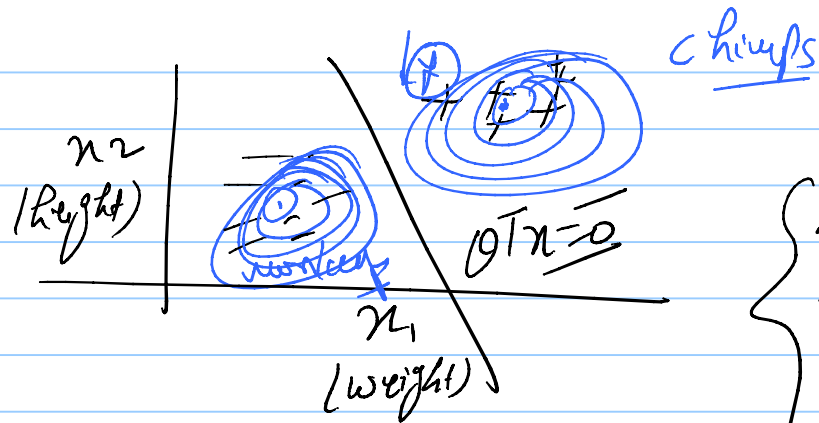
- ① $p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta))$
- ② $\eta = \theta^T x$
- ③ $\eta_0(x) = E[y|x; \theta]$
→

by
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Section 9
Linear & Logistic Reg.

9.3
Softmax Regression

GDA :- Gaussian Discriminant Analysis :-



$$\{x^{(i)}, y^{(i)}\}_{i=1}^m$$

$$x^{(i)} \in \mathbb{R}^n$$

$$y^{(i)} \in \{0, 1\}$$

$$y^{(i)} \in \{1, \dots, C\}$$

$$\begin{cases} \lambda_2(\theta) = \log \left(\prod_{i=1}^m P(y^{(i)} | x^{(i)}, \theta) \right) \\ \hookrightarrow \text{Logistic Regression} \\ P(y=1|x;\theta) = \frac{1}{1+e^{-\theta^T x}} \end{cases}$$

Generative model

Discriminative model

$$P(x^{(i)}, y^{(i)}; \theta) \Rightarrow$$

$$P(y|x; \theta) \stackrel{2}{=} \frac{P(x|y; \theta) P(y; \theta)}{\sum_x P(x, y; \theta)} \quad \underbrace{P(x; \theta)}_{\text{generative}}$$

$$y = \bar{1}$$

$$\begin{aligned} y &\sim \text{Multinoulli}(\Phi) \quad \Phi_c = 1 \\ &\hookrightarrow \Phi = (\Phi_1, \dots, \Phi_C) \\ &\sim \text{Bernoulli}(\Phi); \quad y \in \{0, 1\} \\ x|y=1 &\sim N(\mu_1, \Sigma_1) \end{aligned}$$

Multivariate
Normal Distribution $X \sim \mathcal{N}$

$(X_1 \dots X_n)$

$$P(X_1 \dots X_n) = \prod_{j=1}^n P(X_j)$$

in case

independent $\equiv \text{cov}(X_k, X_j)$

$$\text{cov}(X_j, X_k)$$

$$= E[(X_j - E[X_j])(X_k - E[X_k])] = E[X_j X_k] - E[X_j]E[X_k]$$

$$(X_1 \dots X_n) \sim \mathcal{N}(\mu, \Sigma)$$

$$P(X_1 \dots X_n) =$$

$$\frac{1}{(2\pi)^{n/2}} |\Sigma|^{-1/2} e^{-\frac{(x-\mu)^T \Sigma^{-1} (x-\mu)}{2}}$$

- determinant function

$$x/y=0 \sim \mathcal{N}(\mu_0, \Sigma_0)$$

$$X \sim \mathcal{N}(\mu, \Sigma)$$

Co-Variance

symmetric

$$\Sigma = \begin{bmatrix} \dots & 0 \\ \vdots & \ddots \\ 0 & \dots \end{bmatrix}$$

$$\Sigma_{jk} = \text{cov}(X_j, X_k)$$

$$\mu \in \mathbb{R}^n \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix}$$

if X_j 's are independent

$$X \in \mathbb{R} \quad p(x) = \frac{1}{(\sqrt{2\pi}\sigma)^{1/2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (\text{univariate case})$$

$$\Rightarrow \quad \mathcal{L}(\theta) = \log \left[\prod_{i=1}^n p(x_i, y_i; \theta) \right] \quad \theta = \{\Phi, \{a_c, \omega_c\}_{c=1}^C\}$$